

Curricular Design Experiences for Virtual Production, Multi-Institutional Examples, Experiences and Outcomes

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Abstract

Use of virtual production, combining real-world live performance with virtual environments in a real-time implementation [Kadner, 2019], is increasingly common in the production of live-action content. Movies, episodic productions, live broadcasts, commercials and independent productions are leveraging these rapidly developing technologies to improve efficiencies, enhance visual impact and increase creative flexibility while reducing environmental impact of production. [Cabello et al., 2023] Given the expanding implementation of these approaches and industry demand for skills and experience in this space, it becomes valuable to students to experience these approaches within their academic careers. We present academic experiences from multiple institutions and courses that share approaches to presentation, implementation and evaluation of virtual production concepts and techniques, including design considerations and technological implementations, that provide students with the opportunity to learn and experience virtual production within a studio classroom environment.

CCS Concepts

• **Applied computing**; • **Arts and humanities**; • **Media arts**; • **Education**; • **Computing methodologies**; • **Computer graphics**;

Keywords

Virtual Production, Education, Curricular Development

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1 OVERVIEW

We present two different modalities to teaching live-action Virtual Production in higher education through two separate courses: An

ICVFX (In-Camera Visual Effects) Studio Course, offered at Texas A&M University, which introduced students to ICVFX techniques and LED walls for the use of film-making; and the Green Screen Course, offered at Drexel University, which introduced students to green-screen, and live-streamed broadcast scenarios. While each course had unique content, both courses centered around video projects that integrated live performances with virtual computer graphics (CG) environments, and leveraged real-time graphics engines, camera tracking, with additional digital integration technologies, in conjunction with collaborative partnerships with industry professionals.

2 SAMPLE IMPLEMENTATIONS

2.1 The ICVFX Studio Course at Texas A&M

At Texas A&M University, an ICVFX Studio Course was available to both graduate students in the M.S. and M.F.A. in Visualization program and upperclassmen (Juniors and Seniors) from the Bachelor of Science program. The students were not required to have any previous knowledge of ICVFX or film production. However, students came in with an expertise in 3D computer graphics workflows and were familiar with how to construct a cinematic story, work with virtual cameras, create digital assets for virtual worlds, film lighting, and keyframe animation. The learning outcomes for the course were to both showcase the advantages that using ICVFX with an LED wall has for virtual production film projects and start training future possible hires for the industry.

This Course was made possible at the time by a partnership with two industry collaborators, Stray Vista Studios and SISU Cinema Robotics. SISU donated the use of their robot camera arm and an operating crew. Stray Vista, a virtual production studio, provided the LED stage and equipment to film the projects, including cameras and lighting gear. They provided a learning path to get students up-to-speed with working both with a real-time graphics engine (Unreal) and on an LED wall. Stray Vista also provided two mentors throughout the 11-week course, advising students on nearly all aspects of the Virtual Production film-making process. Students meet with these mentors several times a week to receive feedback, listen to lectures, and gain direct insights into the industry, an experience that would be hard to recreate outside of an internship.

The course took place over the summer of 2024. There were 16 students across 3 teams, and each team was tasked with creating a 2-minute film using ICVFX and a large-scale LED wall. Teams started with story development (including script writing and storyboarding), moved on to creating a 3D previsualization of their

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film using Unreal Engine (including building virtual worlds, and animating virtual cameras and actors), filmed on set for 10 days (with each team getting 3 days) and finished with post-production. Each student assumed a key role for their team: Director, Assistant Director, Director of Photography, Production Designer, or Virtual Production Operator, while members of other teams served as crew.

This course is set in the summer due to the high workload and short time frame. However, regular courses could benefit from inviting industry professionals into classrooms for a guest lecture, workshop, or panel discussion. Due to the wide breadth of skills and knowledge needed for a successful ICVFX Virtual Production project, there is an opportunity to invite a variety of professionals with unique skill sets into the classroom. Fostering these industry partnerships is a way to expand upon student workforce development while still maintaining foundational theory, analysis and fundamentals in the classroom. At the time our college did not have a wall or stage; by partnering with an ICVFX studio we were able to give our students much needed training in a professional environment. Other programs could similarly benefit by fostering industry partnerships with ICVFX studios, where in addition to bringing professionals into the classroom, they can also take trips out to the studio to see a professional production.

2.2 The Greenscreen Live-Broadcast Course at Drexel University

The Live Broadcast Greenscreen Virtual Production course was presented at Drexel University's Westphal College of Media, Art & design for undergraduate and graduate students. There were no prerequisite requirements for students, however almost all students participating had an established background in digital media, film & television, or music industry course work.

The primary objective of this class was to engage students in learning to integrate real world production utilizing studio cameras, lighting and live digital video technologies with virtual sets through live greenscreen compositing and camera tracking techniques to create a seamless, live streamed experience. Class sessions included guest talks from industry professionals as well as in-studio demonstrations and practice with technologies and equipment, supported through weekly assignments that included creation of technical documentation of the studio schematics, as well as development of 3D visual components of the digital environments and graphics used in the livestream.

Design considerations such as theme, color, staging, and contextual influences were combined with learning technologies such as genlock video synchronization, greenscreen keying, and SDI video switching were covered along with remote guest integration through Zoom ISO, multi-user live control of video equipment.

The course took place over a 10-week academic quarter, culminating in the execution of a live streamed event featuring Q&A with industry professional guests connected to the broadcast remotely from the east and west coasts of the US as well as Europe in the final, 10th week of the class. A recording of the live broadcast can be viewed at this link: https://youtu.be/MmvdUOGROqk?si=SHSeT_PCLQsi9NIE&t=683

3 Conclusion

While each of these two experiences featured different implementations of virtual production at different institutions, both experiences leaned into involving collaboration with industry professionals, and provided a comprehensive overview of multiple technologies and concepts rather than limiting focus to any singular aspect of production. Further details of these experiences will be presented at the 2025 SIGGRAPH Conference in Vancouver. In their presentation, the authors will share additional details including: (1) the students' experiences; (2) the instructors' impressions of how the course went; (3) what the instructors would change in the course structure to improve the learning experiences; (4) advice on how to conduct courses that achieve similar learning outcomes but with smaller costs or alternative technical setups.

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